

Kello remote API

Revision 1.0

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Document information

Abstract

This document describes the API use to setup kello specific functions.
See also Light Weight Universal Control Interface for the control on the Libre module

Document convention

Revision history

Revision	Date	Description	Author
1.0	22/02/2016	Initial revision	Frederic Germain

Introduction

This document describes the specific API of kello device, accessible through the Libre tunnel.

Message format

Description

The API is REST oriented, methods are called on resource object.

Header	request id	method/response code	resource id	data	tail
'\n' = 0xa					'\r'=0xd
8 bits	8 bits	8 bits	16 bits	x bytes	8 bits

In case of a request/response transaction, the “request id” shall be the same in the request and response.

With this format, data or other field shall not contain '\r' or '\n' characters.

Header	the '\n' character, signal the start of a message
request id	could be anything from 20 to 250 reserved values : - '\r' - '\n' - 255 for broadcast events
method/response code	any ascii characters. general guideline when a resource is defined : - letter are for method or events, with - 'g' for the main GET method - 'p' for a post/put method - 'd' for DELETE method - and more - number are for reply - '0': no error - '1': no data - '2': invalid - '3': no space - '4': busy - '5': not implemented If there are resources that can provide more methods, cleaner to add a letters and corresponding method rather than to create a new resource
resource id	an ascii number, representing a resource
data	variable length data field
tail	'\r' character, end of message

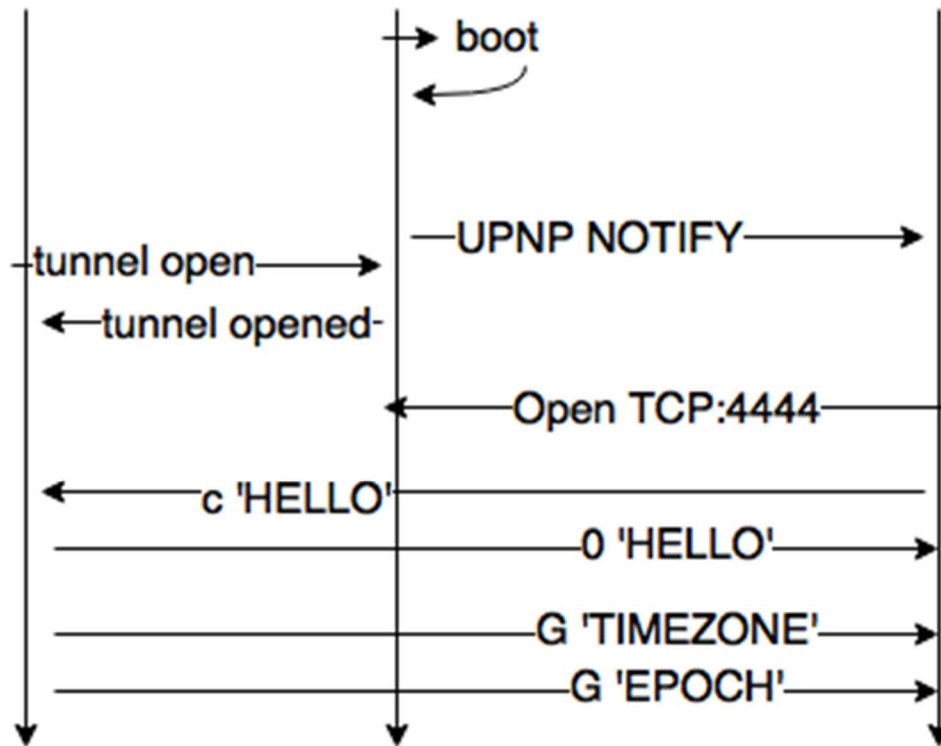
Tunnel interface on libre ls6/ls5b

Upon boot, a tunnel is opened on the libre module, and any device on the wifi network can access it on libre module, TCP port 4444.

Kello module

Libre module

Device



As the TCP socket might not be listening on start, the device shall open try to reopen the socket on start.

Upon opening the connection, the device shall send a 'hello' (00) message.

This might make Kello device request for the timezone and current epoch time.

List of commands

General usage

Undocumented flags shall be set to 0

Generic device API

00: hello

c		send a hello
OK	hello_data	

Used to notify MCU that a device is online, and to get device identification data
hello_data: “%s:%s:%s:%s:%s:%s:%s:%s” % (device_friendly_name, product_id, hardware_version, device_serial, firmware_version, libre_firmware_version, mcu_loader_version, bt_mac_address)

01: locale

g		get current flags
OK	“%x” % (locale_flags)	

p	“%x” % (locale_flags)	set only specified flags
OK	“%x” % (locale_flags)	

d	“%x” % (locale_flags)	clear only specified flags
OK	“%x” % (locale_flags)	flags

screen_flags

1 : display 24h, not 12 am/pm

02: global flags

g		get current flags
OK	"%x" % (global_flags)	

p	"%x" % (global_flags)	set only specified flags
OK	"%x" % (global_flags)	

d	"%x" % (global_flags)	clear only specified flags
OK	"%x" % (global_flags)	flags

global_flags

1 : activity_monitoring_allowed: some user activity are logged by Kello for statistics and future product improvement

2 : crash_report_allowed: crash report are logged by Kello server

04: factory reset

c		Do a factory reset
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05 : manual upgrade

c	"usb"	start an usb upgrade (cf libre docs)
c	"%s" % (xml_upgrade_url)	start internet upgrade with specified url
c		start internet upgrade with latest image

06: screen auto stand by

g		get current flags
OK	"%x" % (screen_flags)	

p	"%x" % (screen_flags)	set only specified flags
OK	"%x" % (screen_flags)	

d	"%x" % (screen_flags)	clear only specified flags
OK	"%x" % (screen_flags)	flags

screen_flags

1 : always on , not auto standby

2 : always off -> not implemented, do not use

07: screen brightness

g		get brightness
OK	"%x" % (screen_brightness)	

p	"%x" % (screen_brightness)	set brightness
OK		

screen_brightness: 1 to 100

08: key emulation

p	key_id	Send a key comand
OK		

This API is for test only, do not use

Key_id: "big_button"|"action"

09: user screen API

For demo and screen definition

p		Show the user screen
d		Delete the user screen
c		Clear the screen
t	"%s" % (text)	Display some text
b	"%x:%x:%x:%x:%s" % (x0,y0,w,h,data)	Display some pixel data

OK		
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This API is for test only, do not use
Key_id: "big_button"| "action"

Time API

10 : epoch time

g		get the epoch time
OK	epoch	

p	epoch	set the epoch time
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epoch : "%x:%x" % (epoch_s, epoch_ns)

11 : timezone

g		get the timezone with offset
OK	timezone_with_offset	

p	timezone_with_offset	set the timezone with offset
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timezone_with_offset: "%x:%x:%x" % (current_offset_seconds, next_change_epoch, next_offset_seconds)

NOTE: add "-" in the beginning of hex integer for negative offsets
2nd & 3rd fields are optional (if no summer time for instance)

19 : gui date

Was coded to shoot the video and have tight control on time display

p		Allow refresh screen every minutes
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d		Disable refresh screen
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Alarm API

20: alarm list

g		get all valid alarm id
OK	“%x” % (alarm_id_mask)	alarm_id_mask: mask of all alarm_id (max 31)

a	alarmfmt	add an alarm
NOSPACE		if no more slot for new alarm is available
NOSPACE	“no_free_spotify_slot”	No place for spotify alarm (might be limited to two)
BUSY		another edit alarm is taking place (setting a spotify alarm could be long)
INVALID	“parse”	request could not be parsed
INVALID	“flags”	invalid flags were detected
INVALID	“uri_too_long”	uri is too long
INVALID	“spotify_not_active”	trying to set a spotify alarm, but spotify is not playing
INVALID	“need_time_set_on_device”	Trying to set a target alarm, but current time has to be set on device
OK	“%x” % (alarm_id)	added at alarm_id

d	“%x” % (alarm_id)	delete an alarm
OK		slot freed

m		get number of alarm that can be set
OK	“%x” % (max_alarm)	

21: edit specific alarm

g	"%x" % (alarm_id)	get an alarm by id
OK	"%x:"alarmfmt % (alarm_id)	

p	alarmfmt	edit a specific alarm
NOSPACE	"no_free_spotify_slot"	No place for spotify alarm (might be limited to two)
BUSY		another edit alarm is taking place (setting a spotify alarm could be long)
INVALID	"id"	invalid id
INVALID	"flags"	invalid flags were detected
INVALID	"uri_too_long"	uri is too long
INVALID	"spotify_not_active"	trying to set a spotify alarm, but spotify is not playing
INVALID	"need_time_set_on_device"	Trying to set a target alarm, but current time has to be set on device
OK	"%x" % (alarm_id)	added at alarm_id

e	"%x:"alarmfmt % (alarm_id)	An alarm setting was updated
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Alarm format

alarmfmt: "%x:%x:%x:%x:%x:%x:%x:%x:%s:%s" % (alam_flags, h, m, days_of_week; volume; target_h, target_m, target_nb_days, target_start_epoch, context_name, uri)

Days_of_week: mask of day of the week this alarm is active. Sunday is first day. For instance, an alarm only for Monday & Tuesday is 0x6.

alarm_flags:

- 1: enabled
- 2: recurrent
- 3: get your back aka important alarm
- 4: target time
- 5: 24h days: takes into account summer/standard time in target time algorithm => for now should be always 0

if it's a non recurring alarm => if enabled, will happen once, then auto-disable

if it's a recurring alarm => days_of_week should not be 0

wakeup program has target time parameters

spotify alarm

spotify use presets to restart audio playlist

to be valid, the player need to be playing spotify during a call to add or to change a spotify playlist.

a change where the playlist name did not change is valid even if spotify is not active. this can be used when the time of alarm or other parameters need to be changed.

When setting an alarm using the spotify, you need to give

"{context_name}:spotify:///blob/{blob}", information you can get with the preset API.

tunein alarm

We use a preset emulation to store alarm on tunein.

When setting an alarm using the tunein, you need to give "{context_name}:tunein:///blob/{blob}".

You can get {context_name} information from the browsing, and you can get the blob with the preset API.

target_* fields

these fields are for alarm with a target. like a wakeup earlier alarm:

I wake up now at 9am. I want to wakeup in 15 days at 8am.

so h=9, m=0, target_h=8, target_m=0, target_nb_days=15, target_start_epoch=1456890737

to know the specifics about how the alarm time are computed based on these parameters, refer to document [Alarm algorithm](#)

pause/start of a alarm with target

An alarm with target can be paused.

When pause, the alarm is inactive, and when reactivated, it will start playing at the time it was supposed to ring next when it was paused.

It implements it easily, this is what is done :

- On pause
 - we override target.h, target.m with the time of next alarm
 - We update remaining days to its current value
- To unpaue
 - Just send all parameter with "enabled" activated and an updated target.start_epoch

22: alarm fsm

g		get an alarm fsm state
OK	“%s:%x:%x:%x” % (alarm_fsm_state, current_alarm_id, next_alarm_epoch, next_alarm_id, current_alarm_start_time, current_alarm_repeat_time)	

State could be :

- "alarms_muted": alarms are forced off by user, alarms_muted flag is on
- "waiting"
- "ringing": alarm is currently ringing (audio may have not started yet)
- "snoozed": alarm snoozed the current alarm, alarm going to restart soon

When current_alarm_id or next_alarm_id (hence next_alarm_epoch) is not defined, no data is returned.

current_alarm_start_time, current_alarm_repeat_time are returned depending on alarm_fsm_state.

p	“stop”	stop alarm from ringing
p	“snooze”	send a snooze command
OK		
INVALID		invalid command detected, or bad state

e	“%s:%x:%x:%x” % (alarm_fsm_state, current_alarm_id, next_alarm_epoch, next_alarm_id, current_alarm_start_time, current_alarm_repeat_time)	
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23: general alarm manager settings

g		get current flags
OK	“%x:%x” % (alarm_manager_flags)	

p	"%x" % (alarm_manager_flags)	set only specified flags
OK	"%x" % (alarm_manager_flags)	

d	"%x" % (alarm_manager_flags)	clear only specified flags
OK	"%x" % (alarm_manager_flags)	flags

alarm_manager_flags
1 : alarms_muted

e	":%x" % (alarm_manager_flags)	New settings
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24: auto shutdown alarm setting

g		auto shutdown alarm setting
NO_DATA		Auto shutdown disabled
OK		

p	%x (auto_shutdown_alarm_s)	set only specified flags
INVALID		Invalid digit
OK		

d		Remove auto shutdown feat
OK		

25: snooze duration setting

g		
OK		

p	%x (snooze_duration_s)	
INVALID		Invalid digit
OK		

26: sleep program

Note that get to sleep faster and fall asleep with music program are the same program. The only difference is that the display animation flag is at 0 for the fall asleep with music and is at 1 for the get to sleep faster

g	"%x:sleepprog_fmt" % (remaining_secs)	get current sleep program
OK	sleepprog_fmt	

p	sleepprog_fmt	
INVALID		
OK		

d		cancel program, restore volume
INVALID		program was not started
OK		

sleepprog_fmt: "%x:%x:%x:%x:%x:%s:%s" % (remaining_duration_secs,
sleep_program_duration_secs, sleepprog_flags, screen_brightness_100, volume,
context_name, uri)
sleepprog_flags
1: display animation

2: play content

27: get your back settings aka important alarm

Important alarm is just a regular with the get your back flags

g		get get your back settings
NO_DATA		not set
OK	"%x" % (get_your_back_flags)	

p	"%x" % (get_your_back_flags)	set get your back settings
OK		

d		delete get your back settings
OK		

get_your_back_flags

1: play_sound

2: double tap

3: remote_snooze_only

4: iftt_get_your_back

28: snooze less program

g		get current program
NO_DATA		not set
OK	"%x:%x:%x:%x" % (snooze_less_flags, target, days_of_week, reset_day_of_week)	

p	"%x:%x:%x:%x" % (snooze_less_flags, target, days_of_week)	set current program
OK		

d		remove program
OK		

reset_day_of_week: first day of week when we should reset the snooze count. Sunday is 0.

snooze_less_flags

1: play_sound

2: double tap

3: remote_snooze_only

4: iftt_harder_snooze

29: bedtime alert

g		get current program
NO_DATA		not set
OK	"%x:%x:%x" % (bedtime_alert_flags, h,m)	

p	"%x:%x:%x" % (bedtime_alert_flags, h,m)	set current program
OK		

d		remove program
OK		

bedtime_alert_flags

1: display star sky

Account API

TBD

Player API

40: volume

g		get volume
OK	"%X" % (volume)	

p	"%X" % (volume)	set volume
OK		

volume 0...100

41: player state

p	"uri:%s" % (content_uri)	load the playlist and play (spotify not supported)
p	"alarm:{alarm_id}"	Load a content associated with an alarm and play it
OK		

d		stop the player
OK		

42: content browser

l	content_directory	
OK	content_directory_listing	

m		Get default alarm content
OK	alarm_content_object	

content_directory: “/wakeuptones/local”

content_directory_listing : json list of alarm content object

alarm_content_object: json object with fields :

- “uri”: string of the alarm content
- “title”: English title for the track

We provide a default alarm, which is stored into the device, and that we play in case there is any problem streaming the selected content in time, or in case of error.

The default alarm is the 1st entry of the list.

45: presets

g	“%s” % (music_provider)	get current preset data - if no music_provider is given, then the current music source
OK	preset_data	

s	“%s:%s” % (music_provider,context_name)	Set tune-in preset
s	“%s” % (music_provider)	Set spotify preset
OK	preset_data	
INVALID	bad_music_provider	
NOSPACE		Could be returned

The preset API depends on the music provider

Spotify

Use spotify connect.

preset_data: “%s:%s:%s” % (music_provider, blob, context_name)

music_provider: "spotify"
blob: opaque blob
context_name: name of the playlist

Note on format spotify blob :

4 bytes	16 bytes	4 bytes	x bytes
50 00 00 00	playlist id ?	02 73 70 6F	playlist owner

NOTE: Spotify do not have to be saved (SET) on the device. But as tune-in is, you shall use the SET api to have the

Tune-in

GET is not supported

SAVE the preset on the device.

If tunein is not playing during a SAVE, an error INVALID is returned

Note on the tunein blob:

It's actually the radio name for now

URI of content

spotify:///blob/{spotify_blob}	Play the given preset
tunein:///blob/{tunein_blob}	Play the preset radio
deezer:///user/me/flow	Deezer flow
deezer:///user/playlist_by_name/{name of playlist}	
local:///...	Local track as browed via api "content browser"

50: alarm stats

g		get stats
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NO_DATA		time is not set
OK	stats_data	

s	stats_data	get stats
INVALID		error parsing
NO_DATA		time is not set
OK		

d		Clear stats
OK		

formats

stats_data: "%s:%s:%s:%s:%s:%s:%s" % (#daystat[dd], #daystat[d-1], #daystat[d-2], ...)

daystat: "%x,%x,%x,%x" (alarm_time, nb_snoozes, wakeup_time, alarmoff_time)

daystat could be empty if no information available

Each of alarm_time, wakeup_time, alarmoff_time could be empty if no information available

alarm_time, wakeup_time, alarmoff_time: number of seconds since midnight

Maximum is $23 \times 3600 + 59 \times 60 + 59$ (last second before midnight)

Alarm_time is the time at which the alarm was programmed

Wakeup_time is the time when the ringing trigger after last snooze is done. If no snooze was done, this is the alarm time

Alarmoff_time is the time when the user turn off the alarm, or when the auto alarm off trigger.

Test API

95: usb

s	usb_charge_config	Set GPIO on usb charge controller
OK		

Usb_charge_config = (usb_ctl3<<2) | (usb_ctl12<<1) | (usb_en<<0)

String

String (%s) shall all be in utf8

'\', '/', ':' shall be escaped with '\'

About volume

There is :

- A main volume. This one is changed with the '40: volume' object, or via libre API.
- One settings by alarm. The device volume is changed to this one when an alarm is active, and restore the main volume when the alarm is over

When want to be able to preview a alarm volume. To do that, the developer shall:

- Get the current volume when entering alarms settings
- Use volumes api when relevant to change the volume
- Restore the volume when exiting the alarms settings

About alarm audio preview

When an alarm is already set, you can use `player_load_alarm (p41alarm:%d)` to start a preview.

To detect the content play, use

- Libre Mb-80 "SUCCESS" for start and mb-80 "NI" for pause/stop
- libre player event mb-51 for other sounds

On new alarms, it depends if contents are presettable or playable by url

- When presettable, suppose the content is always playing and no way to stop (as we stop it at the exit of the alarm setting screen)

Valid for

- Spotify
 - vtuner
- When playable by url, use directly the url given by mcu preset API

About get to sleep program

You cannot delete settings, the 'd' just stop a currently active program

In UI, just show it active when `remaining_duration_secs` is not 0

Otherwise, load settings from kello when starting a new program.